Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/27/2004
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Transgenic Plant Biomonitors of Space Flight Exposure		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Gainesville	**State:**	FL
Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	09/23/2000	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa (University of Florida)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This proposal is designed to be the next-generation of our previous flight project, PGIM-01, and investigates the use of reporter gene technology for the molecular analysis of development and stress in plants within space flight environments. We have designed a series of transgenic plants that carry sensor promoter - reporter gene constructs that are capable of monitoring a variety of environmental and developmental influences that might be altered in microgravity and general space flight environments. The first generation of these experiments with the GUS reporter have been tested for biological veracity on KC-135 parabolic flight missions, and flew as PGIM-01 in the Plant Growth Facility on STS-93. Results of these KC-135 missions and STS-93 were used in this proposal to (1) design a second generation of sensor promoters to refine our definition of the cellular biological impact of microgravity and space flight; (2) design a second generation of reporter gene constructs that include real-time and non- destructive observations of reporter gene activity; (3) design and conduct short and longer term flight experiments that begin a comprehensive		

	analysis of molecular genetic changes induced by space flight; and (4) develop methods to integrate reporter gene activities with data from DNA chip analysis of genome expression. The long-term goal is an understanding of the space flight induced molecular changes in plant gene activity, an understanding that will lead to hardware and/or genetic engineering modifications that will mitigate the negative impacts of space flight on plant growth and development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An understanding of signal transduction in higher plants (and by inference, all eukaryotes) is central to basic molecular biology research.
Task Progress:	In the current period we extended experiments evaluating real-time imaging and data collection of Adh/GFP in the prototype flight hardware being designed at Kennedy Space Center. In the past year we conducted several experiments designed to mimic a long duration shuttle mission (12-15 days) and conducted our Science Verification Test (SVT) with the prototype hardware. Four genetic constructs were used for the SVT and associated trials: CaMV35s/GFP, Adh/GUS, Adh/GFP and RD29/GFP. The RD29/GFP construct is new to the set of transgenic plants being tested in the prototype hardware. RD29 is primarily induced by drought and cold, but not by hypoxia. Work also proceeded with the development of additional transgenes, as well as with the development of more responsive Adh/GFP transgenic lines. A KC-135 experiment was conducted to investigate the effects of parabolic flight on gene expression profiles of the TAGES plants. In addition, the KC-135 flights were used to profile transgene expression and to field test microgravity harvest and stowage techniques and hardware.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Books/Book Chapters	Paul A-L, Ferl RJ. "Gene expression in space biotechnology experiments." in "Plant Biotechnology 2002 and Beyond." Ed. I.K. Vasil. Dordrecht ; Boston : Kluwer Academic Pub., 2003. p. 343-346., Apr-2003
Papers from Meeting Proceedings	Paul A-L, Murdoch T, Ferl E, Levine HG, Ferl RJ. "The TAGES Imaging System: Optimizing a Green Fluorescent Protein Imaging System for Plants." 2003 International Conference on Environmental Systems, Vancouver, BC, Canada, July 2003. 2003 International Conference on Environmental Systems, Vancouver, BC, Canada, July 2003. SAE technical paper 2003-01-2477. http://dx.doi.org/10.4271/2003-01-2477 , Jul-2003